

Multimodal human-state estimation under uncertainty: a survey aligned to AHSEF

A structured survey of six shortlisted papers, read against the AHSEF pipeline (Audio + Video + Speech → HSEN → uncertainty → HSIG → UGAPR), the ten datasets held locally, and an RTX 3050 / RTX 4000-series training budget. Every quantitative value is taken directly from the source PDFs.

PAPERS SURVEYED	MODALITY SCOPE	DATASETS HELD
6 (of 8 shortlisted)	Audio + Video + Speech/Text	10 local corpora

TARGET
First ~50% of AHSEF

01 Scope, method and coverage

This is a targeted survey, not a broad review. The corpus is the pre-shortlisted set; no additional literature search was performed. Each paper is read for six things only: its architecture, the datasets and modalities it evidences, its strongest quantitative results, its most transferable ablation, its computational cost, and the distance between what it solves and what AHSEF must solve.

AXIS	WHAT WAS EXTRACTED	WHY IT DECIDES RELEVANCE FOR AHSEF
Modality coverage	Which of audio / video / speech-text the paper actually evidences	AHSEF's first milestone is trimodal; audio-only or physiology-only evidence does not transfer directly
Dataset overlap	Overlap with the ten corpora on disk	Reproducibility — we can only re-run what we hold
Pipeline stage	Mapping onto HSEN / uncertainty / HSIG / UGAPR	Identifies which stage each paper can actually populate
Quantitative strength	Reported metric, protocol, and margin over the best baseline	Distinguishes a real advance from a protocol artefact
Compute	Params, GPU memory, training time, hardware	Hard filter: RTX 3050 has 4 GB, the workstation card is the ceiling
Reusability	REUSE / ADAPT / BASELINE / REFERENCE ONLY	Converts the survey into an implementation decision

Coverage of the pipeline by the surveyed literature

AHSEF STAGE	HUSFORMER	EMOTION2VEC	MPLMM	LDDU	ESED	AVT-CA	COVERAGE
Modality encoders	—	●	○	○	●	●	Well covered
Multimodal fusion (HSEN)	●	—	●	●	●	●	Well covered
Uncertainty estimation	—	—	—	●	●	—	Two papers
Missing / degraded modality	—	—	●	—	○	—	One paper
Information gain prediction (HSIG)	—	—	—	—	—	—	Not covered
Cost-aware routing (UGAPR)	—	—	—	—	—	—	Not covered
Continuous valence / arousal	○	—	—	—	—	—	Not covered

● direct contribution · ◦ partial or incidental · – absent. Husformer classifies valence and arousal into three coarse levels on DEAP, which is not continuous regression. The bottom three rows are the survey's central finding.

Excluded, with reason. Two shortlisted items were not supplied as PDFs — the Transformer Autoencoder for incomplete-data multimodal emotion recognition, and the Active Inference / EFE-guided micro-gesture paper. They are omitted entirely rather than summarised from memory, and no figures are quoted for them anywhere in this report. The Active Inference paper is the only shortlisted item that addresses information-seeking observation, so its absence leaves the HSIG/UGAPR row above with no coverage at all; that row should be re-checked once the PDF is available.

02 Dataset landscape — what we hold and what the literature uses

Ten corpora are on disk. The table below maps each to the supervision it actually provides, the AHSEF component it can train, and which surveyed papers report results on it. Sizes are quoted where a surveyed paper states them.

DATASET	MODALITIES	SUPERVISION IT PROVIDES	ROLE IN AHSEF	SURVEYED PAPERS USING IT
CMU-MOSEI	A + V + T	7-way emotion (multi-label) and sentiment; 44 977 utterances / 91.9 h / ~1 000 speakers [e2v T1]; intensity levels quantised 0 / 0.3 / 0.6 / 1.0	Primary. HSEN training, unaligned fusion, missing-modality sweep, HSIG counterfactuals	LDDU, MPLMM, emotion2vec, AVT-CA
IEMOCAP	A + V + T	6-way emotion (5 810 train / 1 623 test utterances [ESED T1]); dimensional valence / arousal / dominance annotations	Primary. HSEN training, the only held corpus with both categorical and dimensional labels on all three modalities	ESED, emotion2vec, MPLMM
MELD	A + V + T	7-way emotion, conversational context; 9 989 / 1 109 / 2 610 utterances [ESED T1]; heavy class imbalance	Primary. Conversational generalisation, temporal modelling, class-imbalance stress test	ESED, emotion2vec
CREMA-D	A + V	6-way acted emotion, 7 442 clips from 91 actors, four intensity levels, separate audio-only / video-only / audiovisual ratings [AVT-CA §V]	Controlled A/V ablation. The per-presentation ratings make it the cleanest corpus for isolating modality contribution	AVT-CA
RAVDESS	A + V	8-way acted emotion, 7 360 clips, balanced by class [AVT-CA T1]	Controlled A/V baseline; sanity check for the audio branch	AVT-CA, emotion2vec (82.43 WA)
MSP-Podcast	A (+ T)	Naturalistic speech; categorical emotion plus continuous valence / arousal / dominance	Fills a literature gap. Continuous affect supervision for the audio branch and an out-of-domain generalisation test for emotion2vec	None

DATASET	MODALITIES	SUPERVISION IT PROVIDES	ROLE IN AHSEF	SURVEYED PAPERS USING IT
AffectNet+	V (static)	Facial expression plus continuous valence / arousal in the wild; soft / ambiguity-aware labelling	Fills a literature gap. Pre-training the face encoder and the only held source of continuous facial v/a	None
FERPlus	V (static)	10 crowd annotations per image, retained as a <i>label distribution</i> rather than a single class	Fills a literature gap. The annotator distribution is measured aleatoric uncertainty — a direct target for calibrating the HSEN uncertainty head	None
RAF-DB	V (static)	Basic and compound facial expressions, crowd-labelled with per-image distributions	Face-encoder pre-training; compound labels probe the "blended emotion" ambiguity ESED and LDDU both motivate but never measure	None
WESAD	Physiological	3-state affect / stress from EMG, EDA, BVP, RESP and others	Deferred. Phase-2 physiological extension only	Husformer (78.68 % Acc)

The most consequential observation in this survey. Four of the ten corpora we hold — MSP-Podcast, AffectNet+, FERPlus and RAF-DB — are used by *none* of the six surveyed papers. They are also the only four that supply the two things the surveyed literature demonstrably lacks: **continuous valence/arousal supervision** and **per-sample annotator distributions**.

ESED and LDDU both argue at length that emotional labels are ambiguous and that uncertainty must be modelled — then evaluate on corpora with single hard labels, so their uncertainty is never validated against a ground-truth ambiguity signal. FERPlus and RAF-DB give us exactly that signal. This is a defensible, low-cost contribution available to AHSEF that none of the surveyed work is positioned to make.

Supervision actually available per state dimension

STATE DIMENSION	CORPORA WITH USABLE GROUND TRUTH	VERDICT FOR THE FIRST 50 %
Categorical emotion	IEMOCAP, CMU-MOSEI, MELD, CREMA-D, RAVDESS, FERPlus, RAF-DB	In scope
Valence	IEMOCAP, MSP-Podcast, AffectNet+	In scope
Arousal	IEMOCAP, MSP-Podcast, AffectNet+	In scope
Uncertainty / ambiguity	FERPlus, RAF-DB (annotator distributions); CMU-MOSEI intensity levels as a weak proxy	In scope (novel)
Stress	WESAD only	Defer
Fatigue	None held	Defer
Cognitive load	None held (Husformer uses MOCAS / CogLoad, which we do not have)	Defer
Engagement	None held	Defer

The first implementation should therefore commit to the state vector [valence, arousal, emotion, uncertainty]. The full seven-dimension vector is not defensible on the data we hold, and claiming it

would be unfalsifiable.

ESED: Emotion-Specific Evidence Decomposition for Uncertainty-Aware Multimodal Emotion Recognition in Conversation

Xiong, Ji, Kong, Dai & Yin · CIKM 2025 · doi 10.1145/3746252.3761430

ADAPT

9.5

PURPOSE

Decomposes each modality's *evidence* — not its features — into emotion-consistent, emotion-specific and dynamic components, so that ambiguity, modality conflict and temporal drift become three separately measurable uncertainty sources instead of one confidence scalar.

MODEL / ARCHITECTURE

```
text -> RoBERTa-base (768-d) -+
audio -> WavLM-base (1024-d) -+-> proj d=256 -> e_m = ReLU(f(x))
video -> ViT (768-d) -+ Dirichlet: alpha = e + 1
                                S = sum(alpha), vacuity u = |Y| / S

|
+-- W_cons . e -> emotion-consistent L_align : pull together across
+-- W_spec . e -> emotion-specific L_spec : per-modality cross-er
+-- W_dyn . e -> dynamic L_dyn : 2-layer / 4-head Tran
                                L_orth : pairwise cosine orthogonality between the t

|
v
attention-gated re-projection -> fused Dirichlet -> L_main (CE on expected probability)

L = L_main + 0.2 L_align + 0.3 L_spec + 0.3 L_dyn + 0.2 L_orth
```

DATASET & MODALITY

IEMOCAP (6 classes) and MELD (7 classes), text + audio + visual. Both are AHSEF primaries and both are on disk.

PERFORMANCE

DATASET	METRIC	PROPOSED	BEST BASELINE	BASELINE
IEMOCAP	Weighted F1	72.66	72.54	DQ-Former
MELD	Weighted F1	68.21	66.79	AdaIGN

ESED Tables 2–3. The IEMOCAP margin is only 0.12 pt; ESED wins by being consistently strong across classes (Frustrated 70.15, Excited 84.27, Neutral 73.02) rather than by peaking on any one. On MELD it is the first method in its comparison set to report non-trivial scores on the rare Fear (26.49) and Disgust (28.72) classes.

IMPORTANT ABLATION – MODALITY CONTRIBUTION

MODALITIES	IEMOCAP WF1	MELD WF1	Δ VS TEXT ONLY (IEMOCAP)
Text (T)	69.74	66.28	–
Audio (A)	53.62	52.01	–16.12
Visual (V)	50.13	50.25	–19.61
T + A	71.23	67.34	+1.49
T + V	70.58	66.91	+0.84

MODALITIES	IEMOCAP WF1	MELD WF1	Δ VS TEXT ONLY (IEMOCAP)
A+V	55.77	53.69	-13.97
T+A+V	72.66	68.21	+2.92

ESED Table 6 — the single most decision-relevant table in this survey. Text carries most of the signal; audio and video each add roughly 0.8–1.5 pt on top of it and are close to worthless without it. This is the empirical case for gating: the marginal value of activating A or V is small enough to be worth trading against energy and latency, which is exactly what UGAPR is for.

IMPORTANT ABLATION — ARCHITECTURE AND CAPACITY

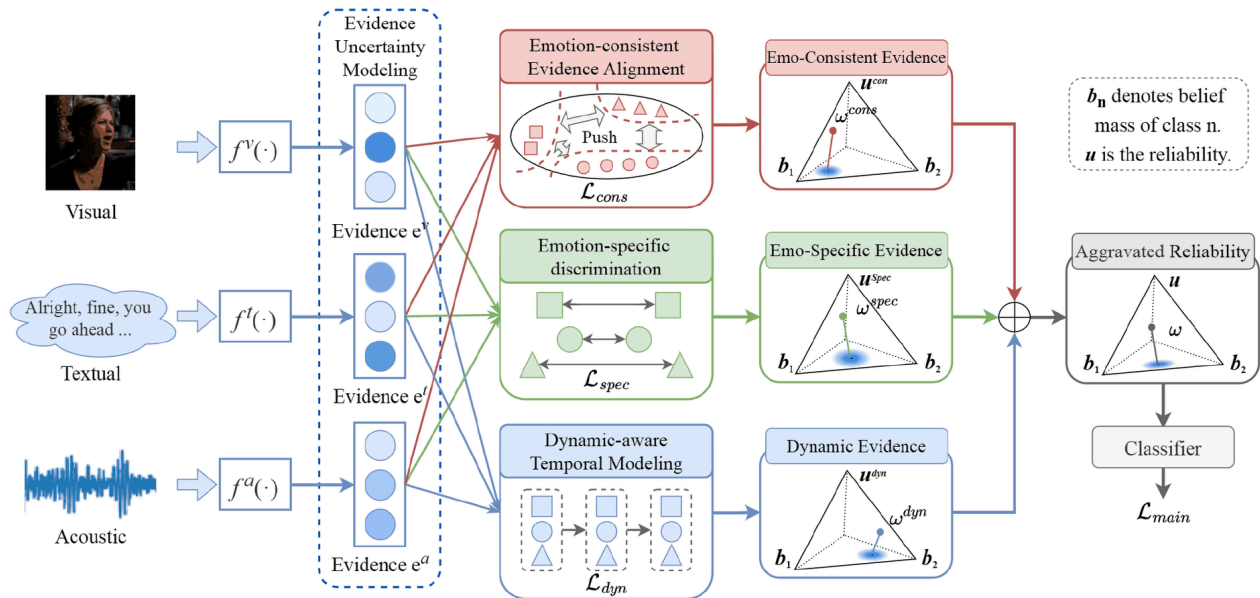
VARIANT	IEMOCAP WF1	MELD WF1	COST OF REMOVAL
Full ESED	72.66	68.21	—
w/o emotion-consistent evidence	69.84	65.92	-2.82 / -2.29
w/o emotion-specific evidence	71.37	67.10	-1.29 / -1.11
w/o dynamic evidence	71.56	67.32	-1.10 / -0.89
w/o orthogonality loss	70.42	66.75	-2.24 / -1.46

ESED Tables 4–5. Cross-modal consensus matters most. The orthogonality constraint is worth ~2 pt and is a single regularisation term — the cheapest win in the paper.

PROJECTION DIM	ENCODER SIZE	IEMOCAP	MELD
128	Small	67.13	63.57
256	Small	69.54	65.26
512	Small	71.07	65.80
128	Large	71.25	66.12
256	Large	72.66	68.21
512	Large	70.95	67.43

ESED Table 7. Performance peaks at $d = 256$ and *declines* at 512 — the authors attribute this to feature sparsity and over-fitting in low-resource emotional settings. Directly useful: our budget-driven choice of 256 is also the accuracy-optimal one, so there is no compute/quality trade to argue about.

Compute: 2× RTX 3080, PyTorch 1.9, batch 32, Adam lr 1e-4, weight decay 1e-5, linear warm-up 10 %, max 50 epochs with early stopping (patience 5), dropout 0.3.



ESED • FIGURE 2 — FRAMEWORK OVERVIEW

The template for AHSEF's uncertainty stage. Each modality produces a Dirichlet evidence vector that is split into three orthogonal subspaces and recombined by *reliability*. Per-modality reliability is precisely the input HSIG needs, which is why this figure — rather than any fusion diagram — is the one that matters most to our design.

USEFUL MODELS / TECHNIQUES

- **Evidential Dirichlet head with vacuity** $u = |Y| / S$ — bounded, per-sample, per-modality uncertainty computed in closed form with no sampling and negligible cost. The single most transferable component in the survey.
- **Orthogonal three-way evidence decomposition** — gives HSIG a *typed* uncertainty (is this ambiguity, conflict, or drift?) rather than one scalar. Different types should drive different routing actions.
- **Per-modality reliability weights** used as the fusion gate — reusable unchanged as UGAPR input features.
- 256-d unified projection before fusion; 2-layer / 4-head temporal Transformer for the dynamic subspace; dropout 0.3.
- Auxiliary-loss weighting is sensitive: (0.2, 0.3, 0.3, 0.2) gives 72.66/68.21, while (0.1, 0.1, 0.1, 0.1) drops to 68.48/64.29 and (0.5, 0.1, 0.1, 0.3) destabilises training. Budget tuning time for this.

GAP VS AHSEF

- Uncertainty steers *fusion weights*, never *acquisition*. Every modality is always encoded and always present at inference; there is no notion of cost.
- No missing-modality evaluation. Table 6 is a training-time modality-subset study, not a test-time dropout or degradation study.
- Categorical emotion only — no valence/arousal regression, and no calibration metric (ECE, reliability diagram) is reported, so the uncertainty is never shown to be trustworthy in the sense HSIG requires.
- Conversation-level model: the dynamic subspace assumes dialogue turns, which AHSEF may not have at inference time.

CONCLUSION FOR AHSEF

ADAPT · 9.5/10. Adopt the evidential head, the orthogonality loss and the reliability gate; drop the conversational assumption and add the calibration evidence the paper omits. This is the closest thing in the shortlist to a reference implementation of HSEN plus uncertainty.

emotion2vec: Self-Supervised Pre-Training for Speech Emotion Representation

Ma, Zheng, Ye, Li, Gao, Zhang & Chen · Findings of ACL 2024, pp. 15747–15760

REUSE

9.5

PURPOSE

A universal, frozen speech-emotion representation: pre-trained by online teacher/student distillation on 262 h of unlabelled emotion audio, then used downstream with nothing but a linear probe.

MODEL / ARCHITECTURE

```
raw audio -> CNN feature extractor -> 12-layer Transformer backbone    (93.79 M params, FROZED)

pre-training: teacher/student online distillation, both initialised from data2vec 2.0
  loss = frame-level masked-prediction loss + utterance-level loss (chunk embedding)
  corpus = LibriSpeech-960 init + 262 h of open-source emotion audio

downstream: linear layers only    (0.20 M trainable params)
```

DATASET & MODALITY

Speech only. Reports on IEMOCAP, MELD, RAVDESS, CMU-MOSI and CMU-MOSEI — four of which we hold — plus nine non-English corpora.

PERFORMANCE

DATASET	METRIC	PROPOSED	BEST BASELINE	BASELINE
IEMOCAP (linear probe, 5-fold LOSO)	WA %	71.79	68.58	data2vec 2.0 (base)
IEMOCAP vs SER specialists	WA %	71.79	71.80	DST (WavLM-large + 22.78 M trainable)
IEMOCAP vs large SSL	WA %	71.79	70.03	WavLM-large (316.62 M)
MELD	WA / UA / WF1 %	51.88 / 28.03 / 48.70	48.92 / 26.10 / 45.80	data2vec 2.0
RAVDESS	WA / WF1 %	82.43 / 82.39	81.04 / 80.97	data2vec 2.0
CMU-MOSEI (sentiment)	WF1 %	76.56	74.75	Whisper encoder
CMU-MOSI (sentiment)	WF1 %	69.16	65.41	Whisper encoder

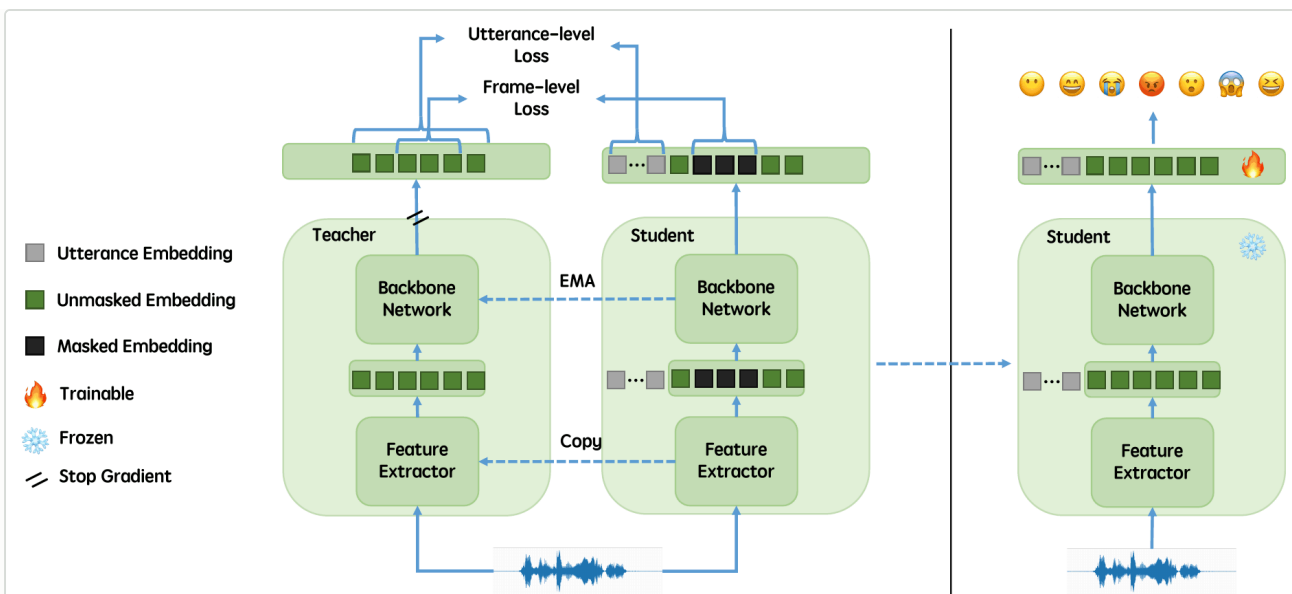
emotion2vec Tables 2, 3 and 7. The decisive row is the first: a *linear* probe on frozen 93.79 M-parameter features matches a fine-tuned WavLM-large specialist while training 0.20 M parameters instead of 22.78 M. Under a same-fold protocol the paper also reports 74.48 % and, with leave-one-speaker-out 10-fold, 72.94 / 77.64 % — quote the 71.79 figure, which uses the stricter protocol.

IMPORTANT ABLATION — SPEECH ALONE VS SPEECH WITH TEXT

MODALITY	MODEL	UAR %	MACRO-F1 %
Speech	Shi et al. 2023	65.01	65.91
Speech	emotion2vec	77.19	76.71
Text (reference)	Shi et al. 2023	77.30	76.67
Speech + Text	Shi et al. 2023	80.18	80.01
Speech + Text	emotion2vec + BERT	81.68	80.75

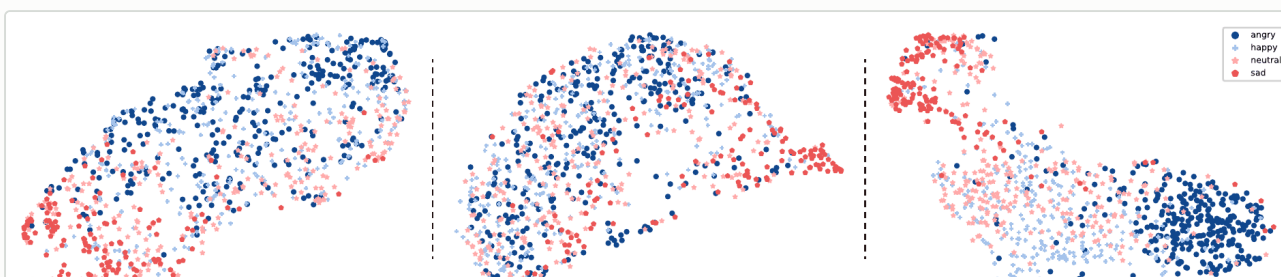
emotion2vec Table 6 — emotion prediction in conversation on IEMOCAP, 10-fold leave-one-speaker-out, six turns of history. Two readings matter for AHSEF. First, a strong audio encoder closes almost the entire audio-to-text gap on its own (77.19 vs 77.30 UAR). Second, adding text on top is still worth +4.5 UAR. This partially contradicts ESED Table 6, where audio alone reached only 53.62 WF1 — the difference is encoder quality, not the modality. Our audio branch choice therefore changes the economics of routing.

Pre-training ablations (Tables 8–11), relevant only if we ever re-pretrain — we should not: frame-level loss is load-bearing (utterance-loss-only collapses to 28.96 WA); the two losses at 1:1 weighting is optimal; chunk-based utterance embedding beats token and global variants; initialising from data2vec 2.0 beats a cold start by 10.5 pt (71.79 vs 61.34).



EMOTION2VEC • FIGURE 1 — PRE-TRAINING FRAMEWORK AND DOWNSTREAM USE

The right-hand panel is our deployment mode: the backbone is frozen (snowflake) and only a lightweight downstream head is trained (flame). This is what makes a 93.79 M-parameter audio encoder viable on a 4 GB laptop GPU — we pay for it once, at feature-extraction time, and never again.



EMOTION2VEC • FIGURE 3 — AROUSAL STRUCTURE IN THE FROZEN FEATURES

Left to right: WavLM, data2vec, emotion2vec, on IEMOCAP, with red and blue marking low and high arousal. Why a linear probe suffices: WavLM and data2vec features show heavy overlap between high and low arousal, whereas emotion2vec's separate cleanly along a continuous high-to-low arousal gradient. Since AHSEF needs continuous arousal, a representation that is already ordered along that axis matters more than raw classification accuracy.

USEFUL MODELS / TECHNIQUES

- The published checkpoint used **frozen**, with features extracted once and cached to disk. 0.20 M trainable parameters for a strong audio branch — the cheapest large win available to us.
- 768-d frame-level features. Concatenating the utterance embedding gives no benefit (71.37 vs 71.79), so use frame features only and skip the extra plumbing.
- For sentiment-style targets, the mean of the last four layers (their CMU-MOSI/MOSEI setting).
- Demonstrated cross-lingual and cross-domain generalisation over nine languages, which supports transferring it to MSP-Podcast without re-pretraining.

GAP VS AHSEF

- Audio only — no fusion architecture, no visual branch, and nothing about modality interaction beyond the single speech+text concatenation in Table 6.
- No uncertainty of any kind; a linear probe yields an uncalibrated softmax, so nothing here feeds HSIG.

- Categorical and sentiment targets only — no valence/arousal regression and no CCC, despite the UMAP evidence that arousal is linearly recoverable.
- No latency, throughput or energy figures, which is what UGAPR will ultimately be scored on.

CONCLUSION FOR AHSEF

REUSE · 9.5/10. Freeze it, cache the features, spend the entire trainable budget on fusion, HSEN and HSIG. No cheaper credible alternative appears anywhere in this shortlist.

LDDU: Latent Distribution Decoupling — A Probabilistic Framework for Uncertainty-Aware Multimodal Emotion Recognition

Huang, Zhong, Lei, Gao, Yang, Wang, Li & Wei · Findings of ACL 2025, pp. 24123–24138

ADAPT

9.0

PURPOSE

Models aleatoric uncertainty by representing every (emotion, modality) pair as a Gaussian in a latent emotion space — the mean carries semantics, the variance carries uncertainty — then fuses the two according to how uncertain the sample is, with an explicit calibration constraint tying variance to error.

MODEL / ARCHITECTURE

```
v: MTCNN facial keypoints -+
a: COVAREP                --> n_m Transformer layers per modality -> O_m
t: BERT                    -+

      |
      Emotion Space Modelling: trainable emotion embeddings  $L = [l_1..l_q]$ , Q-Former-style attn
      ->  $Z_m$  in  $R^{\{q \times d\}}$  (label-related, modality-specific)
      |
      MLP + two FC heads ->  $N(\mu_i^m, \sigma_i^m)$  per label  $i$ , per modality  $m$ 
      |
      SupCon contrastive decoupling over  $3 \times q$  distributions, queue size 8192
      similarity uses BOTH normalised mean and normalised variance
      |
      uncertainty-aware gated late fusion:  $y = d * y_{\mu} + (1 - d) * y_{\sigma}$ 
       $d = \text{error of the auxiliary info-classifier}$ 
      |
      ordinality calibration  $L_{crl}$ :  $\text{rank}(\text{variance norm } S, \text{distance } D) \sim \text{rank}(\text{prediction error})$ 
```

DATASET & MODALITY

CMU-MOSEI (multi-label, 6 emotions) and M³ED — audio + visual + text, evaluated in both aligned and unaligned settings. CMU-MOSEI is an AHSEF primary; M³ED is not held.

PERFORMANCE

DATASET / SETTING	METRIC	PROPOSED	BEST BASELINE	BASELINE
CMU-MOSEI, aligned	Acc / miF1	0.494 / 0.587	0.494 / 0.581	CARAT
CMU-MOSEI, unaligned	Acc / miF1	0.496 / 0.587	0.466 / 0.544	CARAT
M ³ ED	Acc / miF1	0.690 / 0.807	0.664 / 0.788	CARAT
CMU-MOSEI, unaligned	miF1	0.587	0.355 / 0.196	Qwen2-VL-7B / GPT-4o

LDDU Tables 1–2. The unaligned row is the one that transfers: +3.0 accuracy and +4.3 miF1 over CARAT *without any CTC alignment module*, because emotion-space modelling extracts label-related features directly from sequences of differing length. A 6.1 M-parameter model also beats GPT-4o on this task by 39 miF1 points, which is worth remembering before anyone proposes an MLLM shortcut for AHSEF.

IMPORTANT ABLATION – WHICH COMPONENTS CARRY THE UNCERTAINTY MODELLING

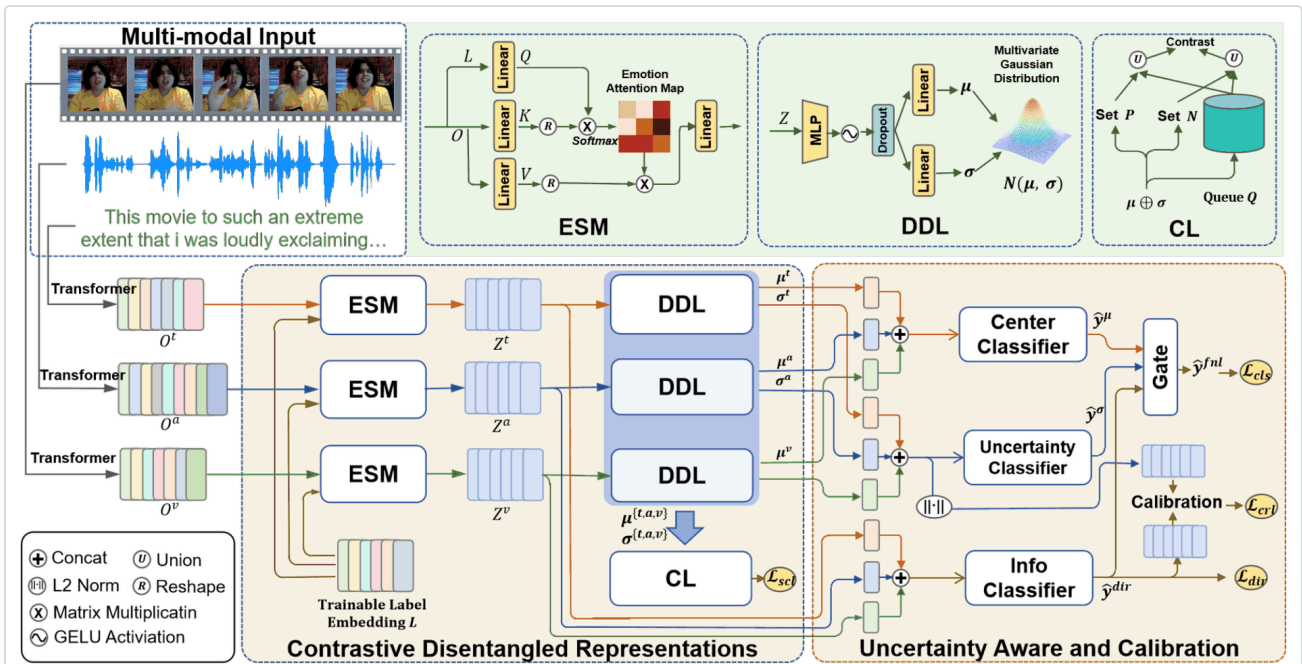
VARIANT (ALIGNED CMU-MOSEI)	ACC	MIF1
Full LDDU	0.494	0.587
w/o contrastive decoupling L_{scl}	0.483	0.575
w/o calibration loss L_{crl}	0.483	0.580
w/o emotion space modelling	0.478	0.577
w/o variance in the similarity	0.483	0.578
w/o centre in the similarity	0.492	0.581
w/o contrastive queue	0.487	0.578
w/o all three main components	0.462	0.561

LDDU Table 3. Author-stated contribution rank: $L_{\text{scl}} > L_{\text{crl}} > \text{ESM}$. Both mean *and* variance must enter the similarity computation — a variance-blind decoupling loses 0.9 miF1, which is the paper's evidence that the variance is carrying real information rather than being a decorative second head.

COMPUTATIONAL EFFICIENCY

MODEL	TOTAL TIME	GPU MEMORY	PARAMETERS
LDDU	505 s	7 872 MB	6.1 M (with CL)
TAILOR	1 315 s	8 730 MB	10.9 M
CARAT	1 318 s	11 262 MB	5.4 M (with CL)

LDDU Table 4 — single A6000, batch 64, 20 epochs, contrastive queue 8 192. Main experiments use batch 128, lr 2e-5, 30 epochs. 7.9 GB exceeds the RTX 3050's 4 GB, but at batch 16–32 with mixed precision and a shorter queue this is laptop-trainable; it fits the 4000-series workstation card comfortably. Note that CARAT has fewer parameters yet costs 43 % more memory and 2.6× the time — parameter count alone is a poor proxy for our hardware constraint.



LDDU · FIGURE 2 – FRAMEWORK OVERVIEW

Three components AHSEF wants at the HSEN output: emotion-space modelling (ESM) to pull label-relevant features out of unaligned sequences, distribution decoupling (DDL) to produce a mean and a variance per emotion, and the uncertainty-aware fusion + calibration block on the right that turns the variance into a signal HSIG can trust rather than merely read.

USEFUL MODELS / TECHNIQUES

- **μ/σ split per emotion per modality** — a natural fit for AHSEF's [v, a, emotion, uncertainty] state vector, where σ becomes the uncertainty channel directly.
- **Ordinality calibration loss:** a rank-agreement constraint between predicted variance and actual error. Cheap to implement, and it is what makes the uncertainty *usable* by a downstream policy rather than merely present. This is the technique with the highest ratio of value to implementation cost in the survey.
- Uncertainty-gated late fusion $y = d \cdot y_\mu + (1 - d) \cdot y_\sigma$, with d estimated by an auxiliary classifier's error.
- Native handling of unaligned streams without CTC — matches AHSEF's asynchronous sensor reality and removes an entire preprocessing stage.

GAP VS AHSEF

- Aleatoric uncertainty only. There is no epistemic term, so the question HSIG must answer — "would acquiring another modality reduce my uncertainty?" — cannot be answered from σ alone.
- No missing-modality or degraded-input experiments whatsoever; all three modalities are always present.
- Multi-label categorical output; no continuous valence/arousal and no CCC.
- The authors explicitly note the model ignores CMU-MOSEI's emotion-intensity labels (0 / 0.3 / 0.6 / 1.0) — that is precisely the supervision AHSEF wants for continuous affect, and it is sitting unused in a dataset we already hold.

CONCLUSION FOR AHSEF

ADAPT · 9.0/10. Take the calibration loss and the μ/σ output form; use the unaligned CMU-MOSEI result (0.496 / 0.587) as the external number our HSEN must approach before we add routing.

Multimodal Prompt Learning with Missing Modalities for Sentiment Analysis and Emotion Recognition (MPLMM)

Guo, Jin & Zhao · ACL 2024 (Long Papers), pp. 1726–1736

BASELINE

9.0

PURPOSE

Handles missing modalities by freezing a backbone pre-trained on a high-resource dataset and training only three families of prompts — a parameter-efficient alternative to architectural repair networks, with prompt count linear rather than exponential in the number of modalities.

MODEL / ARCHITECTURE

```
pre-train Mult backbone on CMU-MOSEI (high-resource) -> FREEZE all backbone parameters

train only:
  P_G generative prompts + Missing Modality Generation Module -> synthesise absent modalities
  P_MS missing-signal prompts -> inform the model THAT a modality is absent (inserted at the end)
  P_MT missing-type prompts -> inform the model WHICH modality is absent (inserted at the end)
  + Conv layers in MMGM + output head

prompt length l_p = 16 ; training-time modality dropout rate eta = 70%
trainable parameters = 5-10% of backbone ; prompts alone = 0.5-1% and constant in backbone
```

DATASET & MODALITY

CMU-MOSEI (used for backbone pre-training), CMU-MOSI, IEMOCAP and CH-SIMS — audio, visual and text across all six non-empty modality subsets. Three of the four are held locally.

PERFORMANCE — PER MISSING-MODALITY CASE

IEMOCAP · ACC %	{A}	{V}	{T}	{A,V}	{A,T}	{V,T}	AVG.
Lower bound (no handling)	46.35	48.07	56.06	58.12	72.18	65.63	57.74
Modality dropout	48.22	48.26	61.26	58.08	72.40	67.08	59.22
MCTN	51.62	45.73	63.78	55.84	69.46	68.34	59.19
MMIN	59.00	51.60	68.02	65.43	75.14	73.61	65.47
MPMM (prompt baseline)	58.69	55.18	68.39	63.68	74.90	73.80	65.77
MPLMM (proposed)	59.77	57.61	69.23	67.26	75.98	74.68	67.42

MPLMM Table 1. Averages on the other datasets: CMU-MOSI 72.14 ACC / 72.57 F1 (MMIN 70.84 / 71.22), CMU-MOSEI 73.75 / 74.68 (MMIN 73.37 / 74.39), CH-SIMS 72.07 / 77.42 (MMIN 71.41 / 76.89). Read the columns as a modality-value table for AHSEF: on IEMOCAP, {a,t} reaches 75.98 while {a,v} reaches only 67.26 — dropping video costs about 1.3 pt, dropping text costs about 8.7 pt.

BACKBONE (CMU-MOSI)	COMPLETE	COMPLETE + PROMPTS	INCOMPLETE	INCOMPLETE + PROMPTS
MISA	80.8	81.4 (+0.6)	67.9	73.4 (+5.5)
MMIM	84.1	84.9 (+0.8)	68.6	72.3 (+3.7)
UniMSE	86.9	87.4 (+0.5)	69.8	75.1 (+5.3)

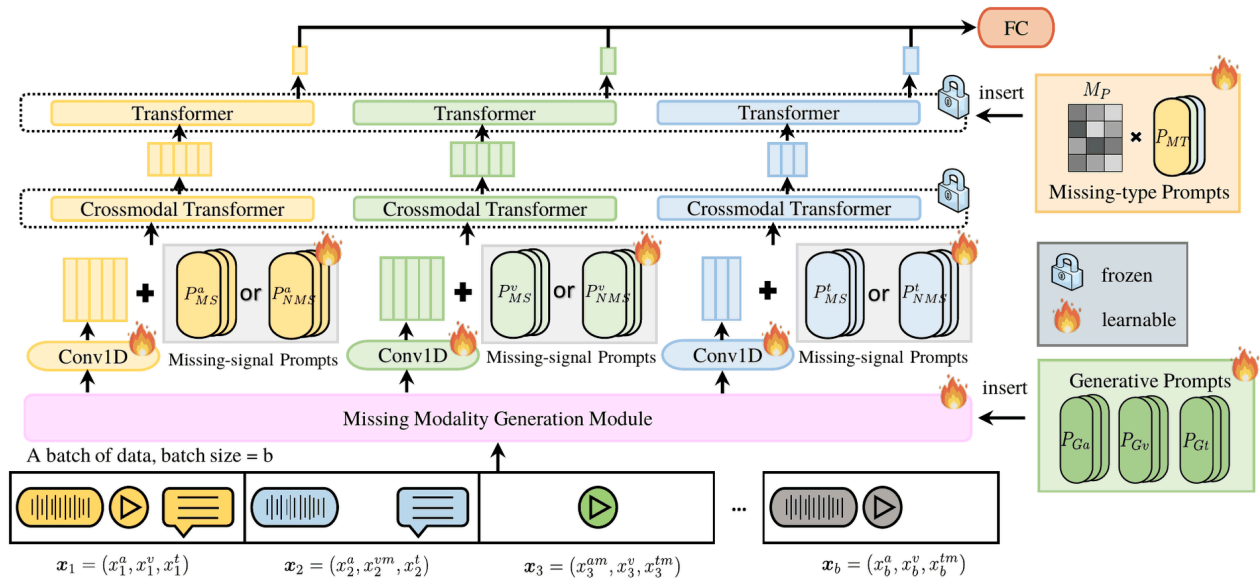
MPLMM Table 2. This verifies the 69.8 → 75.1 figure carried in the pre-analysis brief: it is UniMSE on incomplete CMU-MOSI data, averaged over six missing conditions. The prompts are backbone-agnostic and help slightly even on complete data, which suggests the missing-signal and missing-type prompts teach genuine intra- and inter-modality structure rather than only patching absences.

IMPORTANT ABLATION — WHICH PROMPTS MATTER

P _G	P _{MS}	P _{MT}	MOSI ACC	MOSI ACC-7	MOSI CORR	IEMOCAP ACC
–	–	–	65.04	27.41	0.549	59.22
✓	–	–	70.26	28.27	0.574	64.12
–	✓	–	67.62	28.11	0.580	62.27
–	–	✓	66.41	30.17	0.561	63.48
✓	✓	–	71.94	30.63	0.576	65.23
✓	–	✓	71.60	31.27	0.583	66.11
–	✓	✓	67.56	32.88	0.593	64.92

MPLMM Table 3, averaged over the six missing-modality cases. Generative prompts dominate accuracy and F1; missing-signal prompts help correlation most; missing-type prompts help fine-grained 7-class accuracy most. The three are complementary rather than redundant, which is why AHSEF should treat "which modality is off" as explicit model input rather than an implicit consequence of zeroed features.

Compute — the most encouraging number in the survey for our hardware: all experiments ran on **one 10 GB RTX 3080** with batch size 64, 30 epochs, Adam at lr 1e-3, prompt length 16. Trainable parameters are 5–10 % of the backbone, and the prompts themselves 0.5–1 %, a fraction that does not grow when the backbone does.



MPLMM • FIGURE 1 – ARCHITECTURE

Note where the prompts attach: missing-signal prompts at the Conv1D stage and missing-type prompts at the Transformer stage, around a fully frozen backbone. AHSEF's UGAPR produces exactly this kind of "which modalities are active" signal, so this insertion pattern is the natural way to feed a routing decision back into the estimator.

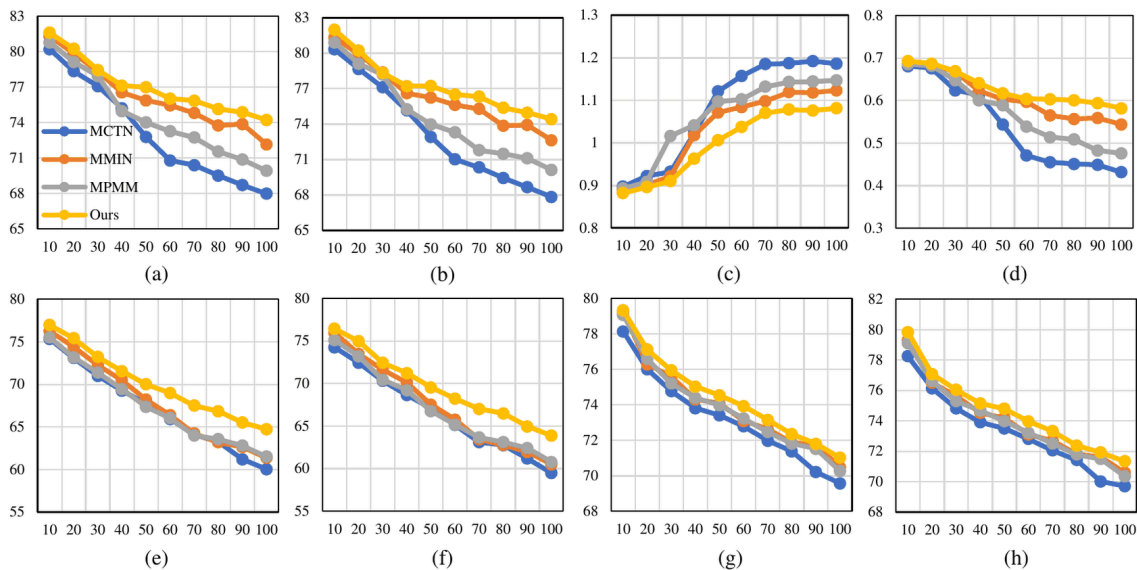


Figure 4: Performance comparison with different modality missing rates during tests. (a): ACC on CMU-MOSI. (b): F1 score on CMU-MOSI (c): MAE on CMU-MOSI. (d): Corr on CMU-MOSI. (e): ACC on IEMOCAP. (f): F1

MPLMM • FIGURE 4 – PERFORMANCE VERSUS MODALITY MISSING RATE

The experiment shape AHSEF needs: performance as a continuous function of modality availability, not a single complete-versus-missing pair. Despite training at a fixed 70 % missing rate, the method degrades gracefully across 10–100 % at test time. We should reproduce these axes with *activation rate* substituted for *missing rate* — the same curve, reinterpreted as a cost/quality operating characteristic.

GAP VS AHSEF

- Missing modality is exogenous — equipment failure, corruption, privacy. It is never a decision the system makes in order to save energy or latency.
- No uncertainty estimate at all, so there is no signal available to drive a policy even if one were wanted.

- No cost, latency or energy accounting; a cheaper modality set is never *preferred*, only *repaired*.
- Requires a high-resource pre-training stage on CMU-MOSEI before prompts can be learned, which adds a dependency to our schedule.

CONCLUSION FOR AHSEF

BASELINE, with protocol adoption · 9.0/10. Its η -masking training and per-subset evaluation become our HSIG training procedure and headline comparison table. Its generative prompts are also the honest baseline for "route around a deactivated modality" — any UGAPR result must beat prompt-based repair, not just a naive lower bound.

Husformer: A Multi-Modal Transformer for Multi-Modal Human State Recognition

Wang, Jo, Zhao, Wang, Yang, Chen & Min · arXiv:2209.15182v2 (Apr 2023)

ADAPT

8.0

PURPOSE

An end-to-end cross-modal Transformer for human-state recognition covering both affect *and* cognitive load, consuming raw, unaligned, heterogeneous streams without hand-crafted feature engineering or alignment. It is the only surveyed paper framed around "human state" rather than emotion alone, which is why it maps most directly onto HSEN as a concept.

MODEL / ARCHITECTURE

```
X_m (L_m x D_m) -> Conv1D_m -> + positional encoding -> Y_m (L_m x D)      common D for all
concat(Y_1 .. Y_M) = Y_F                                                  low-level fusion

for each modality m:
    Cross-Modal Transformer( target = Y_m , source = Y_F ) -> Z_m          m attends to the
concat(Z_1 .. Z_M) -> Self-Attention Transformer -> Z_F -> FC -> prediction

loss = multi-class focal loss with per-class alpha and focusing parameter gamma
```

The topology choice is the contribution: each modality attends to the *concatenated fusion representation* rather than to every other modality pairwise, so cost grows linearly in the number of modalities instead of quadratically. For a framework that intends to add physiological channels later, that scaling property is worth more than the accuracy numbers.

DATASET & MODALITY

DEAP and WESAD (affect), MOCAS and CogLoad (cognitive load) — EEG, EMG, EOG, GSR, BVP, RESP, HR, eye-blink. **No audio, video or text anywhere in the paper.** Of our ten corpora it touches only WESAD, which we have deferred.

PERFORMANCE

DATASET	METRIC	PROPOSED	BEST BASELINE	BEST SINGLE MODALITY
WESAD	Acc / F1 %	78.68 / 79.51	65.76 / 66.32	64.09 (RESP)
Raw DEAP — valence	Acc / F1 %	79.64 / 79.87	65.68 / 66.39	72.80 (EEG)
Raw DEAP — arousal	Acc / F1 %	79.94 / 80.44	66.31 / 66.39	73.27 (EEG)
Preprocessed DEAP — arousal	Acc / F1 %	91.33 / 91.35	86.55 / 87.13	84.86 (EEG)
Raw MOCAS	Acc / F1 %	87.37 / 87.47	75.33 / 75.44	68.25 (EEG POW)

DATASET	METRIC	PROPOSED	BEST BASELINE	BEST SINGLE MODALITY
CogLoad	Acc / F1 %	74.06 / 74.93	61.44 / 63.39	56.45 (GSR)

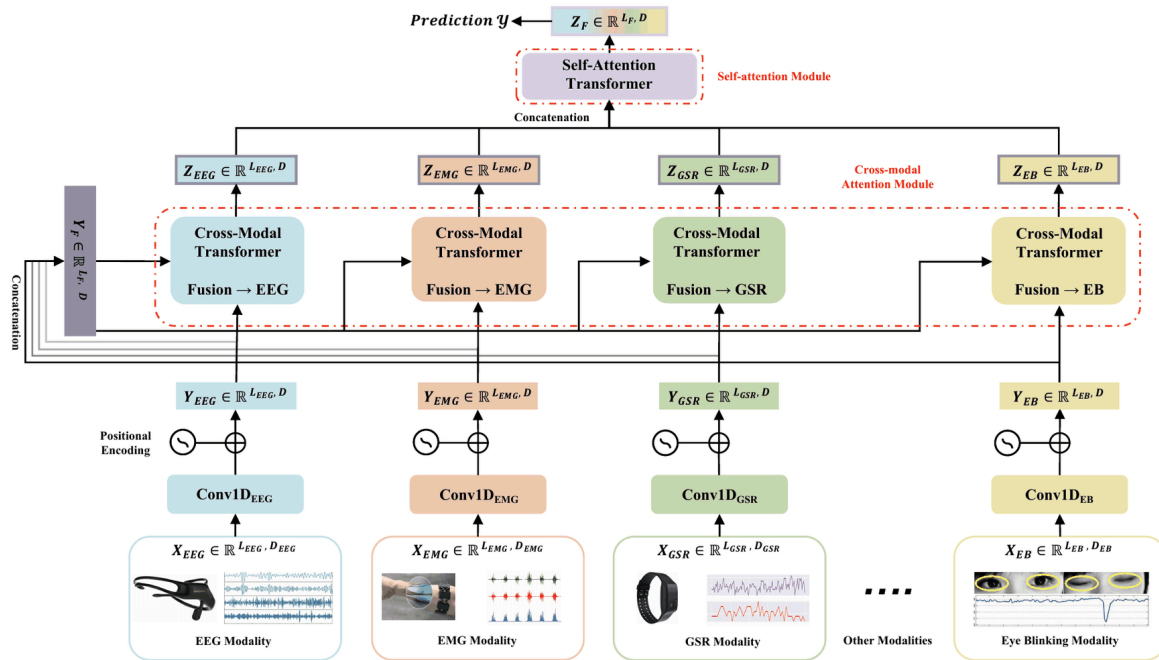
Husformer Tables V, VI and IX–XI. The best baseline is MMResLSTM throughout. The multimodal-over-unimodal margin is widest exactly where single-modality signal is weakest — +14.6 pt on WESAD, +17.6 pt on CogLoad — and narrowest where one modality already dominates (+6.8 pt on raw DEAP, where EEG alone reaches 72.80). That relationship is the general principle AHSEF’s routing exploits.

IMPORTANT ABLATION — WHAT THE FUSION STRATEGY IS WORTH AND WHAT IT COSTS

VARIANT	WESAD ACC	RAW MOCAS ACC	COGLOAD ACC	PARAMS	GPU MEM (WESAD)
HusFuse — concatenation + self-attention only	68.77	70.65	58.49	—	—
HusLSTM — LSTM replaces self-attention	70.64	78.98	67.09	—	—
HusPair — pairwise cross-modal attention	73.57	82.12	65.11	3.90 M	5 417 MB
Husformer — fusion-to-modality attention	78.68	87.37	74.06	0.71 M	3 084 MB

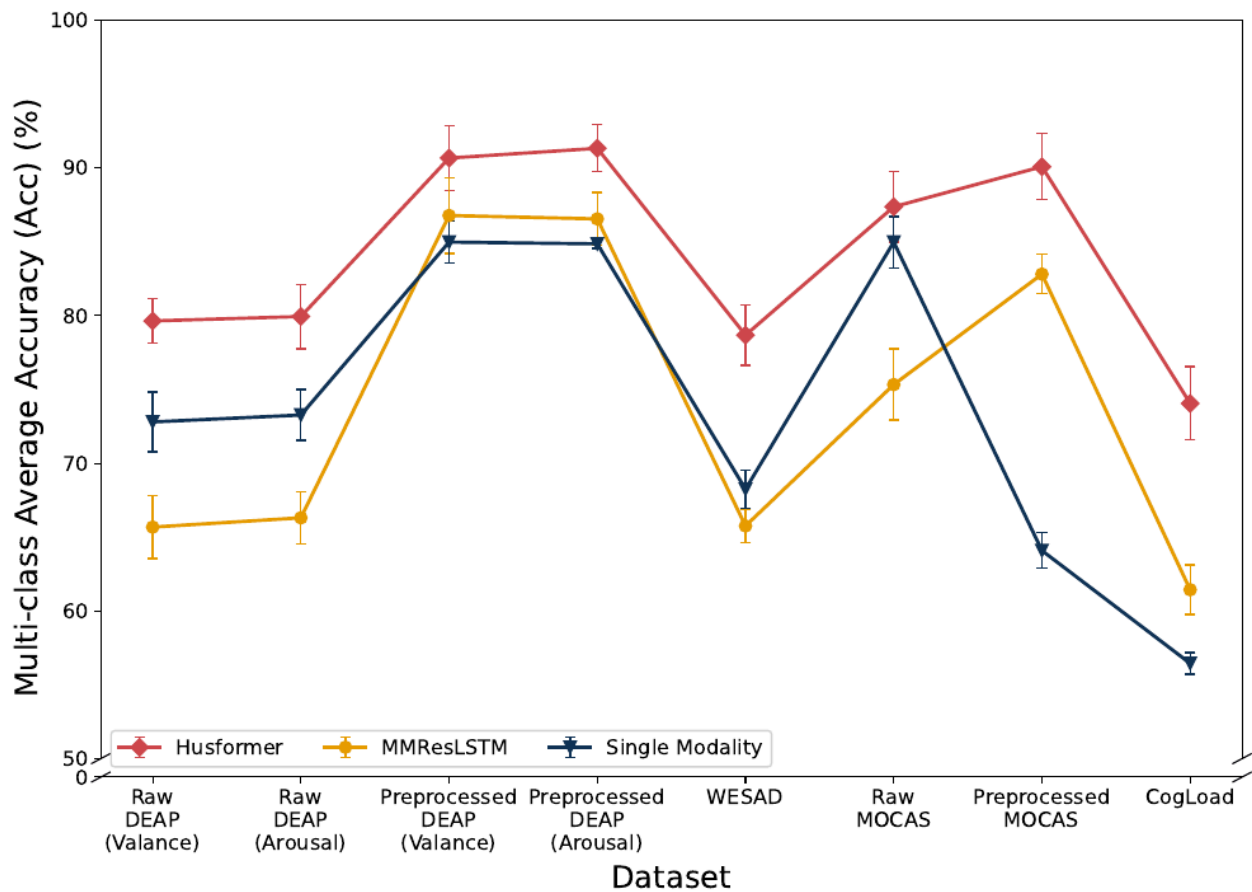
Husformer Tables V–VII. The headline efficiency result: attending to a single concatenated fusion representation beats full pairwise cross-attention by 5.1 pt on WESAD while using **5.5× fewer parameters** and **43 % less GPU memory**. Across all six dataset settings Husformer stays within 0.63–0.75 M parameters and 94–3 084 MB. Removing cross-modal attention entirely (HusFuse) costs 9.9 pt on WESAD, so the mechanism — not merely the depth — is doing the work.

Hyper-parameters actually used: Transformer hidden size 40, 3–5 cross-modal attention heads, cross-modal block dropout 0.05–0.1, output dropout 0.1, Adam at 1e-3 to 2e-3, 20–80 epochs, batch 64–1024 depending on dataset. A remarkably small model by current standards, and the authors note HusPair’s parameter count alone limits its practical applicability — the same argument AHSEF makes about always-on sensing.



HUSFORMER • FIGURE 1 – FRAMEWORK

The HSEN topology we should adopt. Substitute the four physiological branches for emotion2vec, a face encoder and RoBERTa, and keep both the fusion-to-modality cross-attention and the single self-attention stage. The Conv1D + positional-encoding front end is also what lets modalities of different sampling rates and lengths meet at a common dimension without explicit alignment.



HUSFORMER • FIGURE 3 – MULTIMODAL FUSION VERSUS THE BEST SINGLE MODALITY

The margin this architecture buys over the best available unimodal predictor. For AHSEF this gap is the budget HSI_G is allowed to spend: any modality UGAP_R chooses not to activate must cost less accuracy than the margin shown here, or the routing is not worth having.

USEFUL MODELS / TECHNIQUES

- **Fusion-to-modality cross-attention** — linear rather than quadratic in the modality count, with a clean parameter-count justification. The core reusable idea.
- Conv1D front end plus positional encoding to bring heterogeneous, unaligned sequences to a common dimension D without CTC or resampling.
- Multi-class focal loss with per-class α — directly applicable to the class imbalance we will hit on MELD and IEMOCAP.
- A sub-1 M-parameter fusion trunk, which leaves the entire RTX 3050 budget for the encoders and the routing components.

GAP VS AHSEF

- No audio, video or speech evaluation. The claim that the topology transfers to A/V/T is untested and is ours to verify — this is the single largest assumption in our proposed stack.
- No uncertainty output. Predictions are bare softmax over three coarse levels, so nothing feeds HSI_G.
- No missing-modality or degraded-input experiments; all modalities are assumed present and healthy.
- Three-level classification of valence and arousal, not continuous regression, so no CCC and no comparability with dimensional affect literature.

CONCLUSION FOR AHSEF

ADAPT · 8.0/10. Adopt the topology and the efficiency argument; ignore the datasets entirely. Its priority sits below the A/V/T papers only because none of its evidence is in our modalities — the architecture itself remains our first choice for the HSEN trunk.

AVT-CA: Audio-Video Transformer Fusion with Cross Attention

arXiv:2407.18552v4

REFERENCE ONLY

5.5

PURPOSE

An audio–video cross-attention Transformer aimed at temporal misalignment between the two streams, with an attention-refined video front end. Despite evaluating on CMU-MOSEI it uses **no text branch**.

MODEL / ARCHITECTURE

```

audio -> 2x [ Conv1D -> BatchNorm -> ReLU -> MaxPool ] -----+
                                                                +-> Transformer blocks -> self-attention
video -> Conv2D -> channel attention (x) spatial attention |
                    -> local patch feature extractor
                    -> inverted residual blocks (x2) -----+
                        |
                    bidirectional cross-attention: audio -> video and video -> audio (4 heads)
                        |
                    max-pool each stream -> element-wise ADD -> fully connected -> softmax

```

DATASET & MODALITY

RAVDESS (7 360 clips, 8 classes), CMU-MOSEI (23 454 clips reduced to 6 balanced emotion classes), CREMA-D (7 440 clips, 6 classes) — audio + video only. Three of our datasets, but only two of our three modalities.

PERFORMANCE, AS REPORTED

DATASET	PROPOSED ACC / F1 %	BEST BASELINE	BASELINE
RAVDESS	96.11 / 93.78	95.95 / 92.17	CNN
CMU-MOSEI	95.84 / 94.13	84.71 / 84.51	MAG-BERT
CREMA-D	94.13 / 94.67	83.70 / 79.50	DE III

AVT-CA Tables II–IV.

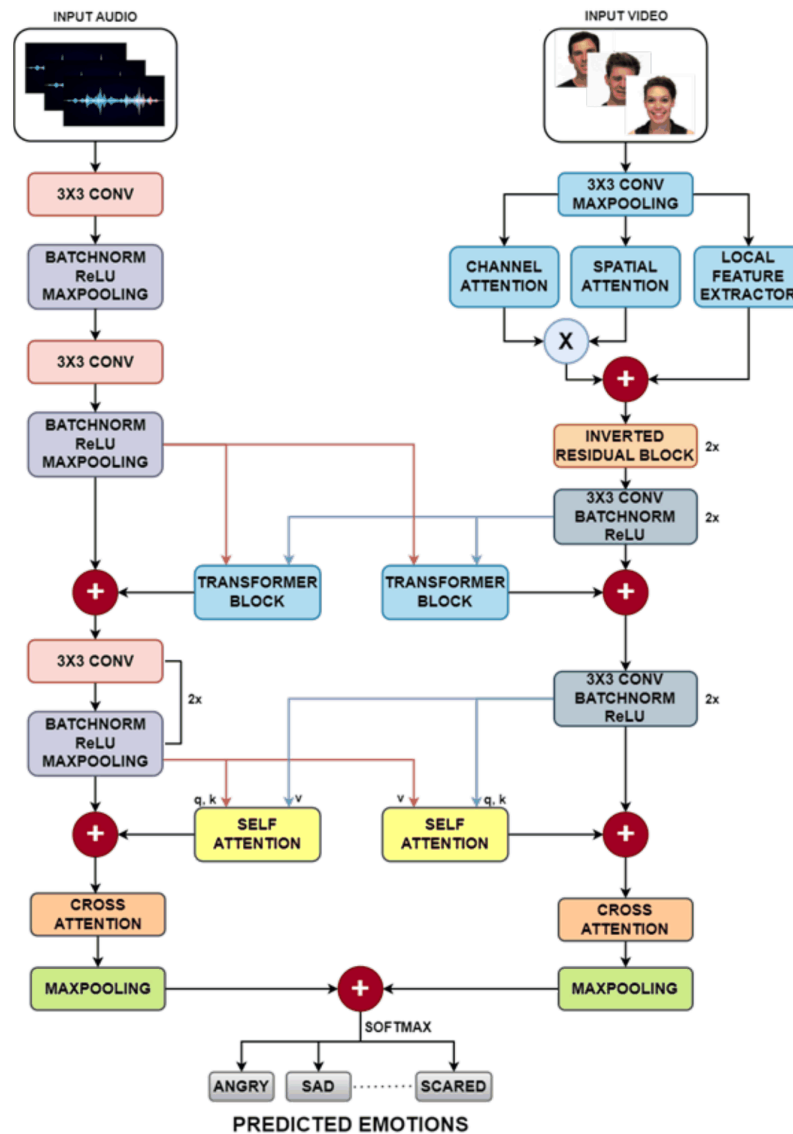
These numbers should not be used as AHSEF targets. The paper reports an 80/20 split with no speaker-independent protocol stated, on a class-balanced six-emotion subset of CMU-MOSEI. Its own listed baselines — MAG-BERT 84.71, MFM 84.41, COGMEN 83.81 — sit where the field sits. An 11-point jump over all of them, from an architecture with no text branch, is far more consistent with speaker leakage across a random utterance-level split than with the fusion mechanism. For comparison, LDDU on the same corpus under a standard protocol reports 0.587 mF1. The ablation reinforces the concern: components scoring in the 60s and 70s individually combine to the mid-90s.

IMPORTANT ABLATION

VARIANT	RAVDESS ACC	CMU-MOSEI ACC	CREMA-D ACC
Intermediate transformer, 1 head (IT-1)	76.41	67.72	71.50
Intermediate transformer, 4 heads (IT-4)	78.50	64.91	74.80
Cross-attention, 1 head (CT-1)	76.00	64.94	72.10
Cross-attention, 4 heads (CT-4)	77.41	67.72	75.50
Full AVT-CA (IT-4 + CT-4)	96.11	95.84	94.13

AVT-CA Table V. A 19–31 pt jump from combining two components that each score in the 60s–70s is not a normal ablation profile. The authors describe it as "approximately 20 % increase" and attribute it to synergy; the more parsimonious explanation is that the full model's training or evaluation configuration differs from the ablated ones.

Compute: approximately 72 h of training on an AMD EPYC 7763 CPU (2 sockets, 64 cores each), batch size 8, Adam at lr 0.01 with weight decay 0.001. No parameter count, FLOPs or inference latency is reported anywhere — which is the one thing an "audio-video efficiency" reference most needed to provide.



AVT-CA • FIGURE 2 – ARCHITECTURE

Included as the contrast case: this is the audio–video-only fusion pattern AHSEF explicitly does *not* adopt. Bidirectional pairwise cross-attention scales quadratically with modality count and offers no natural attachment point for a text branch or an uncertainty head. The channel-plus-spatial attention video front end, however, is a cheap and reusable idea for our own visual encoder.

GAP VS AHSEF

- No text or speech branch, so it cannot represent AHSEF's strongest modality — the one ESED Table 6 shows carries most of the signal.
- No uncertainty, no missing-modality handling, no cost model.
- No model size, latency or FLOP figures, so it cannot serve as an efficiency reference despite being the most "edge-flavoured" paper in the set.
- Evaluation protocol is not stated to be speaker-independent, so its numbers are not comparable with ours or with the rest of the survey.

CONCLUSION FOR AHSEF

REFERENCE ONLY • 5.5/10. Keep the channel/spatial-attention video front end as a cheap visual encoder trick and cite the paper for the temporal-misalignment framing. Do not use it as a baseline and do not treat its accuracy figures as targets.

04 Cross-paper synthesis

4.1 What the evidence says about each modality

FINDING	EVIDENCE	SOURCE	CONSEQUENCE FOR AHSEF
Text is the strongest single modality in conversational emotion	69.74 vs 53.62 (A) and 50.13 (V) WF1	ESED Table 6, IEMOCAP	Text should be the always-on modality; A and V are the gated ones
Losing text is far more damaging than losing video	$\{a, t\}$ 75.98 vs $\{a, v\}$ 67.26 ACC	MPLMM Table 1, IEMOCAP	Confirms the same ordering under a completely different method and protocol
Audio's weakness is an encoder problem, not a modality problem	53.62 WF1 (WavLM-base) vs 77.19 UAR (emotion2vec)	ESED Table 6 vs emotion2vec Table 6	A strong audio encoder changes the routing economics — do not fix the gating policy before fixing the encoder
Video adds least and costs most to compute	+0.84 WF1 over text alone	ESED Table 6, IEMOCAP	Video is the first modality UGAPR should learn to switch off
Full trimodal fusion still wins	72.66 vs 71.23 (best bimodal)	ESED Table 6	The always-on ceiling is real but narrow — routing has ~1.4 pt of headroom to protect, not 10
Fusion gains are largest when no modality dominates	+14.6 pt (WESAD), +6.8 pt (raw DEAP)	Husformer Fig. 3 / Tables IX–XI	Gating should be state-dependent: activate more modalities precisely when no single one is confident

Two independent papers, two methods, two protocols, one ordering: text > audio > video. The third row is the important caveat — that ordering was measured with a weak audio encoder, and our stack does not use one.

4.2 How the two uncertainty papers differ

AXIS	ESED	LDDU	WHAT AHSEF SHOULD TAKE
Formalism	Evidential deep learning, Dirichlet over classes	Gaussian per (emotion, modality) in a latent space	Dirichlet — bounded, closed-form, no sampling
Uncertainty quantity	Vacuity $u = Y / S$, per modality	Variance σ , per emotion per modality	Both: vacuity for routing, σ for the state vector
Decomposition	Three typed subspaces (consistent / specific / dynamic)	None — one variance per pair	ESED's typing, because different types imply different routing actions
Calibration	Not addressed	Ordinality constraint tying variance rank to error rank	LDDU's — ESED's uncertainty is unvalidated without it
Use of uncertainty	Reliability weights in fusion	Gate between μ -prediction and σ -prediction	Neither; AHSEF uses it to decide what to sense
Cost	Negligible over the base model	6.1 M params, 7 872 MB, 505 s / 20 epochs	Both are affordable on the workstation card

The two are complementary rather than competing: ESED supplies a typed, per-modality signal and LDDU supplies the constraint that makes such a signal trustworthy. Combining them is a low-risk design and, as far as this survey can tell, has not been published.

4.3 Computational cost across the survey

PAPER	TRAINABLE PARAMS	GPU MEMORY	TRAINING COST	HARDWARE USED	RTX 3050
Husformer	0.63 – 0.75 M	94 – 3 084 MB	20 – 80 epochs	not stated	Yes
emotion2vec (probe)	0.20 M	not stated	linear probe	not stated	Yes, cached
MPLMM	5–10 % of base	≤ 10 GB	30 ep, batch 64	1× RTX 3080	Batch ↓
LDDU	6.1 M	7 872 MB	505 s / 20 epochs	1× A6000	Batch ↓, AMP
ESED	not stated	not stated	≤ 50 ep, batch 32	2× RTX 3080	Likely
AVT-CA	not reported	not reported	~72 h, batch 8	AMD EPYC 7763 CPU	Unknown

No paper in this survey required more than a single consumer or prosumer GPU, and two of the six ran on RTX 3080-class hardware. Our RTX 4000-series workstation is comfortably sufficient for every method here; the RTX 3050 is sufficient for everything AHSEF actually needs to train, provided encoder features are cached rather than recomputed. Note also that parameter count is a poor memory proxy — CARAT has fewer parameters than LDDU but uses 43 % more memory.

4.4 Comparability of reported results

PAPER	PROTOCOL	SPEAKER-INDEPENDENT	COMPARABLE
emotion2vec	5-fold leave-one-session-out; also 10-fold leave-one-speaker-out	Yes	Yes
ESED	Standard IEMOCAP / MELD splits, weighted F1	Yes	Yes
LDDU	Standard CMU-MOSEI multi-label splits, aligned and unaligned	Yes	Yes
MPLMM	Six missing-modality subsets, five random seeds, η = 70 % training	Yes	Yes
Husformer	Standard splits on physiological corpora	n/a	No – modalities
AVT-CA	80/20 split, protocol not stated, balanced class subset	Not stated	No

Four of the six report under protocols we can replicate exactly. Those four — emotion2vec, ESED, LDDU and MPLMM — are the only external numbers that belong in an AHSEF results table.

05 Priority ranking

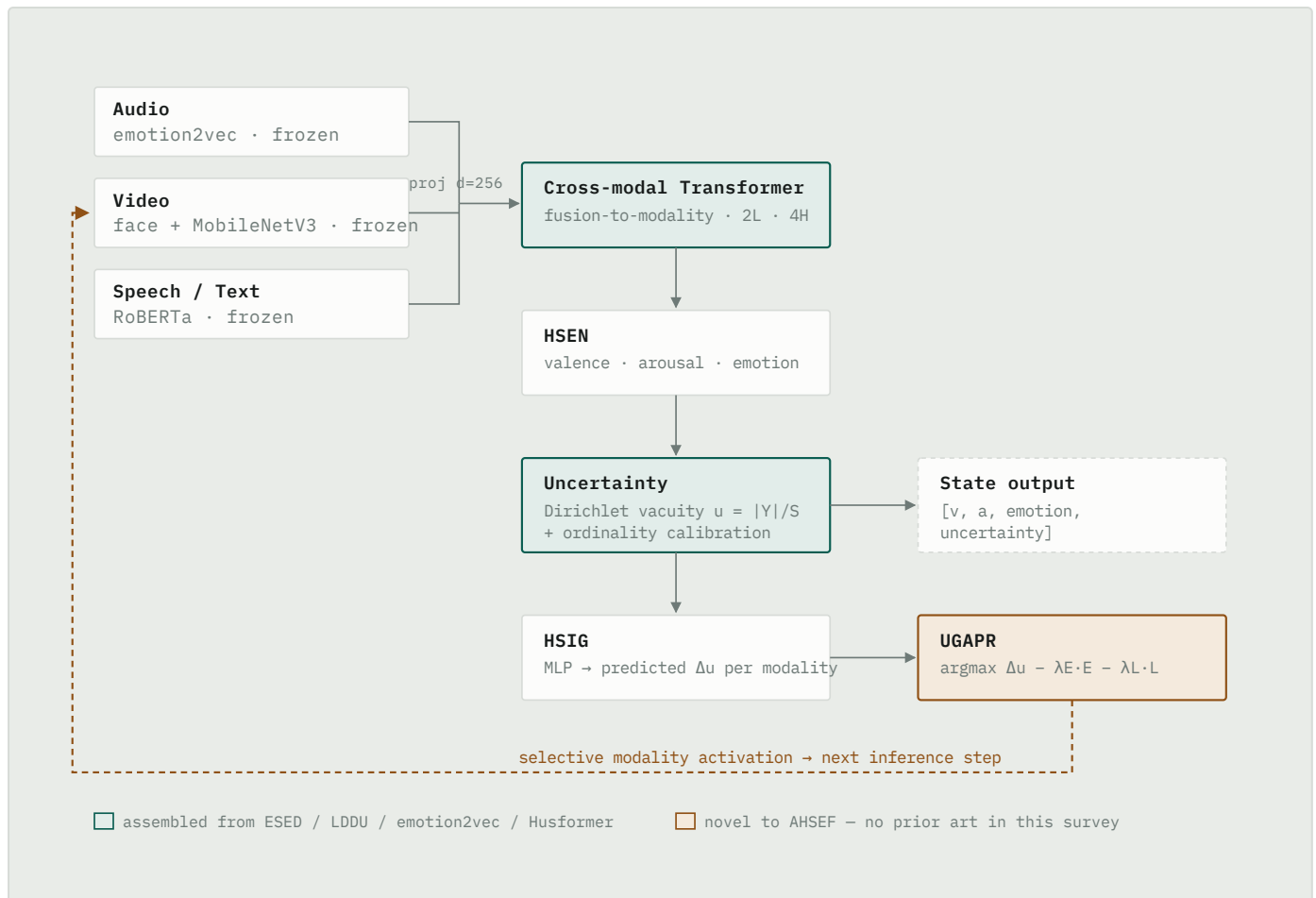
RANK	PAPER	MAIN CONTRIBUTION	DATASET OVERLAP	AHSEF RELEVANCE	REUSE POTENTIAL	PRI.
1	ESED	Evidential decomposition of per-modality uncertainty into consistent / specific / dynamic subspaces	IEMOCAP, MELD	HSEN and uncertainty; supplies the per-modality reliability signal HSIG consumes	Head and losses portable essentially as-is	9.5
2	emotion2vec	Frozen universal speech-emotion representation with linear-probe downstream	IEMOCAP, MELD, RAVDESS, CMU-MOSEI	Audio branch of HSEN	Direct checkpoint reuse, 0.20 M trainable	9.5
3	LDDU	Per-emotion latent Gaussians with ordinality-calibrated uncertainty, unaligned-native	CMU-MOSEI	Uncertainty form and calibration for the HSEN output	Calibration loss and μ/σ head	9.0
4	MPLMM	Prompt-based missing-modality robustness, linear in modality count	CMU-MOSEI, IEMOCAP	HSIG training protocol and the UGAPR evaluation table	Protocol plus generative prompts	9.0
5	Husformer	Fusion-to-modality cross-attention trunk under 1 M parameters	WESAD only (deferred)	HSEN fusion topology and the efficiency argument	Topology yes, weights no	8.0
6	AVT-CA	Bidirectional audio ↔ video cross-attention with an attention-refined video front end	CMU-MOSEI, RAVDESS, CREMA-D	Audio-video fusion contrast case	Video front-end idea only	5.5

Ranked by, in order: audio/video/speech coverage → overlap with the ten held corpora → relevance to HSEN, HSIG and UGAPR → strength and comparability of quantitative evidence → feasibility on RTX 3050 / 4000 → usefulness as a runnable baseline. This differs from the pre-analysis brief in two places, both deliberate: Husformer moves from first to fifth because none of its evidence is in our modalities or datasets — the architecture is still our first choice for the trunk — and AVT-CA moves down because its reported results do not survive protocol scrutiny.

06 Recommended model and architecture

COMPONENT	RECOMMENDED MODEL	REASON, FROM THE EVIDENCE ABOVE
Audio	emotion2vec, frozen; 768-d frame features cached to disk	71.79 WA on IEMOCAP with a linear probe — matches a fine-tuned WavLM-large specialist at 0.20 M trainable parameters, and its features are already ordered along the arousal axis
Video	Face detection and crop (MTCNN or RetinaFace) → MobileNetV3 or EfficientNet-B0 per frame → 2-layer temporal Transformer; optional channel + spatial attention front end	ESED Table 6: video is the weakest branch (50.13 IEMOCAP) and adds ~0.84 pt over text. Do not spend a ViT-B budget on it. Pre-train the frame encoder on AffectNet+ / RAF-DB, which we hold
Speech / Text	RoBERTa-base, or DistilRoBERTa if memory-bound; gold transcripts during training, frozen initially	ESED's text encoder; ESED Table 6 and MPLMM Table 1 independently show text is the strongest modality. Use dataset transcripts rather than repeatedly running ASR
Multimodal fusion	Husformer-style fusion-to-modality cross-attention; d = 256, 2 layers, 4 heads, no CTC alignment; focal loss on the categorical head	Husformer: +5.1 pt over pairwise cross-attention at 5.5× fewer parameters. ESED Table 7: d = 256 beats both 128 and 512. LDDU: unaligned streams work without CTC
HSEN	Shared trunk → valence and arousal regression heads + categorical emotion head	Restrict the state vector to [v, a, emotion, uncertainty] — the only dimensions the ten held corpora actually supervise
Uncertainty	ESED evidential Dirichlet head per modality (vacuity $u = Y /S$) with the orthogonality loss, plus LDDU's ordinality calibration loss; validated against FERPlus / RAF-DB annotator distributions	ESED gives per-modality reliability at negligible cost; LDDU's calibration makes that value rank-correlate with error. The FERPlus validation is the piece neither paper could perform
HSIG	2–3 layer MLP predicting expected uncertainty reduction Δu per candidate modality, trained on modality-masking counterfactuals	No surveyed paper does this. Training targets come from MPLMM-style η -masking: mask a modality, measure the actual change in vacuity and error, regress on it
UGAPR	Closed-form: activate $\text{argmax}_m [\Delta \hat{u}_m - \lambda_E \cdot \text{energy}_m - \lambda_L \cdot \text{latency}_m]$, with a vacuity threshold as the stopping rule	Deterministic and inspectable, no reinforcement learning in the first 50 %. Compare against always-on, fixed-order and random routing

Final architecture



Everything shaded green exists in one of the six surveyed papers and can be assembled from published components and checkpoints. Everything shaded amber — the information-gain prediction, the routing decision and the feedback loop — has no precedent in this literature. That is the contribution, and it is also the part with no external baseline to compare against.

07 First 50 % implementation

1

Cache frozen features for CMU-MOSEI and IEMOCAP

emotion2vec 768-d audio, face-crop CNN video, RoBERTa text — extracted once and written to disk. Nothing downstream ever runs an encoder again during training, which is what makes the 4 GB laptop GPU viable for iteration.

2

Train the HSEN trunk on cached features

Husformer topology, $d = 256$, 2 layers, 4 heads, focal loss on the categorical head, mixed precision. Target before adding anything else: ESED-class weighted F1 on IEMOCAP (~72) and LDDU-class miF1 on unaligned CMU-MOSEI (~0.58).

3

Attach the evidential head and prove the uncertainty is calibrated

Dirichlet output with per-modality vacuity, plus LDDU's ordinality calibration loss and ESED's orthogonality loss. Report ECE and the rank correlation between vacuity and actual error, and validate against FERPlus / RAF-DB annotator distributions. A well-calibrated u is a prerequisite for HSIG, not a nice-to-have — if this step fails, everything after it is unfounded.

4

Run the modality-masking sweep

MPLMM protocol: train with $\eta = 70\%$ modality dropout, then evaluate all six modality subsets and a 10–100 % missing-rate curve. This reproduces the ESED Table 6 and MPLMM Figure 4 shapes on our own model, and simultaneously generates the HSIG training set.

5

Train HSIG on the counterfactuals from step 4

For each sample and each currently-inactive modality, the measured change in vacuity and error from the sweep is the regression target. A 2–3 layer MLP over the current state and the per-modality vacuity vector; well under 1 M parameters.

6

Implement UGAPR and run the central experiment

Always-on versus text-only versus fixed-order versus random versus HSIG+UGAPR versus MPLMM-style prompt repair, plotting state-estimation quality against modality activation rate, measured latency and estimated energy. The claim under test: quality within a stated margin of always-on at substantially lower activation.

7

Add continuous affect supervision

Bring in MSP-Podcast (speech valence/arousal) and AffectNet+ (facial valence/arousal) to train and evaluate the regression heads with CCC. This is where AHSEF goes beyond every paper in the survey, all of which stop at categorical output.

8

Generalisation check

MELD for conversational transfer, then CREMA-D and RAVDESS as controlled audio-video ablation sets. Only after the routing result holds on the primaries. WESAD stays out of the first 50 %.

08 Final research gap

GAP	WHAT THE SURVEYED LITERATURE DOES INSTEAD	EVIDENCE
Uncertainty is never converted into an acquisition decision.	ESED and LDDU both estimate high-quality per-modality uncertainty and spend it entirely on fusion weights, with every modality always encoded and always present.	ESED §3.5 gated fusion; LDDU uncertainty-gated late fusion
Missing modality is treated as damage, never as a choice.	MPLMM repairs an absent modality with generative prompts and never asks whether acquiring it was worth its cost. No cost term appears anywhere in the paper.	MPLMM §1 framing — equipment failure, corruption, privacy
No accuracy-versus-activation-cost operating curve exists for audio + video + text.	Efficiency is reported as static parameters, memory and wall-clock (Husformer 0.71 M / 3 084 MB; LDDU 6.1 M / 505 s), never as a quality-versus-sensing-budget trade-off.	Husformer Table VII; LDDU Table 4; AVT-CA reports none at all
Calibrated continuous affect is unaddressed.	All six papers output categorical or multi-label emotion. No CCC, no ECE and no valence/arousal regression appears anywhere in the survey — despite emotion2vec's own UMAP evidence that arousal is linearly recoverable from its features.	LDDU explicitly discards CMU-MOSEI's intensity labels; emotion2vec Fig. 3

GAP	WHAT THE SURVEYED LITERATURE DOES INSTEAD	EVIDENCE
Uncertainty is never validated against measured human ambiguity.	Both uncertainty papers motivate their work with label ambiguity, then evaluate on corpora with single hard labels — so the estimated uncertainty is never checked against a ground-truth ambiguity signal.	ESED §1 (ambiguity, conflict, drift); LDDU Fig. 1 — neither uses annotator distributions

Two consequences follow. First, **the baseline for AHSEF's central claim does not exist and must be constructed by us** — steps 4 through 6 above are not validation work, they are the contribution. Second, **we hold four corpora (MSP-Podcast, AffectNet+, FERPlus, RAF-DB) that none of the surveyed papers use**, and they happen to supply exactly the two things the literature is missing: continuous affect supervision and measured annotator ambiguity. The second gap is cheaper to close than the first and is worth closing early, because a calibrated uncertainty is a precondition for the routing result to mean anything.

Stated as a single testable question: *can HSI_G and UGAP_R hold human-state estimation quality within a small, stated margin of always-on multimodal inference while substantially reducing modality activation — and is the uncertainty driving that decision calibrated against measured human label ambiguity?*

09 Final recommendation

- **Encoders:** emotion2vec (audio, frozen), MobileNetV3 or EfficientNet-B0 face encoder (video, frozen, pre-trained on AffectNet+ / RAF-DB), RoBERTa-base or DistilRoBERTa on gold transcripts (text, frozen). Cache all features; do not fine-tune any encoder in the first 50 %.
- **Fusion:** Husformer fusion-to-modality cross-attention at $d = 256$, 2 layers, 4 heads, no CTC. Not pairwise cross-attention — Husformer shows 5.5× the parameters for 5.1 pt /less accuracy — and not AVT-CA.
- **Uncertainty:** ESED evidential Dirichlet head with per-modality vacuity and the orthogonality loss, plus LDDU's ordinality calibration loss. Treat ECE and vacuity-to-error rank correlation as first-class reported metrics, not diagnostics.
- **State vector:** commit to [valence, arousal, emotion, uncertainty]. Stress, fatigue, cognitive load and engagement have no supervision in the ten corpora we hold and should be deferred with WESAD.
- **Primary datasets:** CMU-MOSEI (unaligned) and IEMOCAP for development; MELD for conversational generalisation; CREMA-D and RAVDESS as controlled audio-video ablation sets only.
- **Under-used datasets to bring in deliberately:** MSP-Podcast and AffectNet+ for continuous valence/arousal, FERPlus and RAF-DB for annotator-distribution supervision of the uncertainty head. None of the surveyed papers use these four, which is the cheapest available novelty.
- **Baselines to run:** always-on multimodal HSEN, text-only, fixed-order routing, random routing, and MPLMM-style prompt repair. External reference points are ESED on IEMOCAP (72.66 WF1) and MELD (68.21 WF1), LDDU on unaligned CMU-MOSEI (0.496 Acc / 0.587 miF1), and emotion2vec on IEMOCAP (71.79 WA).
- **Do not use AVT-CA's figures as targets.** Its CMU-MOSEI 95.84 % is inconsistent with every standard-protocol result in this survey and most likely reflects the split, not the architecture.
- **Core experiment:** state-estimation quality against modality activation rate, latency and estimated energy — the MPLMM Figure 4 axes with activation rate substituted for missing

rate, and always-on as the upper reference line.

- **Metrics:** CCC and MAE/RMSE for continuous affect, weighted F1 for categorical emotion, ECE plus vacuity-error rank correlation for calibration, and activation rate, wall-clock latency and per-decision FLOPs for cost.

SOURCES — AS SUPPLIED

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